

S614APT1PROFESSIONAL TRAINING REPORT

UNIVERSITY PORTAL DESIGNING

Submitted in partial fulfilment of the requirements for the award of
Bachelor of Engineering degree in Computer Science and Engineering with
specialization in Cyber security

by

P.Bhavya

42614102



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

SCHOOL OF COMPUTING

SATHYABAMA

INSTITUTE OF SCIENCE AND TECHNOLOGY

(DEEMED TO BE UNIVERSITY)

CATEGORY - 1 UNIVERSITY BY UGC

Accredited "A++" by NAAC | Approved by AICTE

JEPPIAAR NAGAR, RAJIV GANDHISALAI,

CHENNAI – 600119

OCTOBER 2024

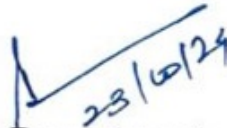
DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

BONAFIDE CERTIFICATE

This is to certify that this Professional Training is the bonafide work of **P. BHAVYA (42614102)** who carried out the project entitled **UNIVERSITY PORTAL DESIGNING** under my supervision from June 2024 to October 2024.



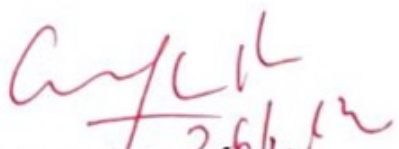
Internal Guide
Dr. R. Shalini M.E, Ph.D


23/10/24

Head of the Department
Dr. A. MARY POSONIA, M.E., Ph.D.,

Submitted for Viva voce Examination held on 26.10.24


Internal Examiner


External Examiner

DECLARATION

I, **P. BHAVYA (42614102)** hereby declare that the Professional Training Report entitled **UNIVERSITY PORTAL DESIGNING** done by me under the guidance of **Dr. R. Shalini M.E, Ph.D** is submitted in partial fulfilment of the requirements for the award of Bachelor of Engineering degree in Computer Science and Engineering with specialization in Cyber Security.

DATE: 28/10/24

PLACE: CHENNAI

P. Bhavya
SIGNATURE OF THE CANDIDATE

ACKNOWLEDGEMENT

I am pleased to acknowledge my sincere thanks to **Board of Management** of **SATHYABAMA** for their kind encouragement in doing this project and for completing it successfully. I am grateful to them.

I convey my thanks to **Dr. T.Sasikala M.E., Ph.D., Dean**, School of Computing, **Dr. A. Mary Posonia M.E., Ph.D., Head**, Department of Computer Science and Engineering for providing me necessary support and details at the right time during the progressive reviews.

I would like to express my sincere and deep sense of gratitude to my Internal Guide **Dr. R. Shalini M.E, Ph.D, Associate Professor**, Department of Computer Science and Engineering for her valuable guidance, suggestions and constant encouragement which paved way for the successful completion of my Professional Training.

I wish to express my thanks to all Teaching and Non-teaching staff members of the **Department of Computer Science and Engineering** who were helpful in many ways for the completion of the project.

COURSE CERTIFICATE



ABSTRACT

This abstract outlines a project aimed at redesigning the university portal to enhance user experience (UX) for all. The existing portal presents challenges, including a navigation structure, outdated design, and limited accessibility, which hinder effective interaction and engagement. By conducting user research, including surveys and focus groups, we will identify and gather insights into user needs and preferences. The redesign will prioritize intuitive navigation, responsive design, and accessibility standards, ensuring a seamless experience across devices. Key features will include personalized dashboards, streamlined information access, and improved integration of academic resources and services. Personalized dashboards for different types of users will be created to provide useful information and tools in one place. For students, this might be: displaying class schedules and deadlines; and for faculty members, class rosters and resources. It will thus lighten up the user's cognitive load because the portal will simplify access to information, so the user can quickly find what he needs without having to navigate through heaps of pages. Again, the seamless integration of available academic resources, including the library, academic advising, and campus services, will make the portal a single stop portal, reducing the requirement to access multiplicity of systems. This holistic approach, when combined with use of responsive design, would ensure consistency and accessibility of the experience to users across devices, including mobile phones and tablets, in addition to usability. By employing user-centred design principles and iterative testing, this project aims to create a more engaging and functional portal that supports the university community's diverse needs. Ultimately, the redesigned portal will facilitate improved communication, enhance academic support, and foster a sense of belonging among users, contributing to a more vibrant and connected university environment.

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	v
	TABLE OF CONTENTS	vi
	LIST OF FIGURES	vii
	LIST OF TABLES	viii
1.	INTRODUCTION	1
2.	LITERATURE SURVEY	9
3.	PROPOSED SYSTEM	12
	3.1 OBJECTIVE	12
	3.2 PROPOSED ARCHITECTURE	13
	3.2.1 DESCRIPTION	14
	3.2.2 REQUIREMENT ANALYSIS	16
	3.2.3 IMPLEMENTATION	17
4.	RESULT ANALYSIS	18
5.	CONCLUSION	19
6.	REFERENCES	20
	APPENDIX	
	I. SCREENSHOTS	21
	II. SOURCE CODE	23

LIST OF FIGURES

Figure No.	Figure Name	Page No.
3.2	Proposed architecture	05
3.2.3	Implementation flowchart	12
4.1	Result analysis ratio	13
6.1	Student login portal	16
6.2	Teacher login portal	17
6.3	Dashboard showing calendar events	18
6.4	Sample navigation map	18
6.5	College login website	19
6.6	Student portal	20
6.7	Teacher portal	20
6.8	About us page	21
6.9	Course Access page	22

CHAPTER 1

INTRODUCTION

University portals play an important role in providing access to academic resources, administrative services, and communication tools for students, faculty, and staff. However, many existing portals are burdened by outdated designs, inefficient navigation, and a lack of user-centric functionality, which can hinder productivity and user satisfaction. As digital platforms have become central to university operations, the demand for well-designed, user-friendly portals is rapidly growing. This report explores key areas for improvement in university portal redesign, focusing on enhancing both user experience (UX) and user interface (UI) to create a more intuitive, personalized, and responsive platform. The redesigned portal will be tailored to different user groups, providing seamless access to essential services with enhanced accessibility, mobile responsiveness, and user-friendly navigation. Features such as integrated course registration, streamlined financial services, improved communication tools, and real-time notifications will be key to creating a powerful and attractive platform. Additionally, the implementation of modern security measures like two-factor authentication and performance optimizations, such as faster load times, will ensure that the portal is both secure and efficient. The goal of the redesign is not only to improve the visual appeal of the portal but also to enhance the core processes, increase work efficiency, and make the platform more accessible to all users. By addressing these areas, the new portal will significantly enhance user engagement, improve overall satisfaction, and streamline daily operations, ultimately supporting the university's digital infrastructure in a more effective and user-friendly way.

CHAPTER 2

LITERATURE SURVEY

University portals are crucial platforms that provide students, faculty, and staff with access to essential academic resources, administrative services, and communication tools. However, many existing portals suffer from outdated designs, inefficient navigation, and a lack of user-centred functionality, resulting in frustration and decreased productivity. As educational institutions increasingly rely on digital infrastructure for daily operations, the demand for modern, user-friendly portals has intensified, prompting a shift toward redesigning these systems with a focus on user experience (UX) and user interface (UI). Research highlights significant challenges in current portals, particularly their reliance on legacy systems that are difficult to navigate and lack personalization; Al-Sharhan and Abbasi (2020) note that cluttered interfaces and poor navigation structures hinder users from quickly finding relevant information. To enhance usability, many institutions are embracing user-centred design (UCD) principles, which prioritize the specific needs and behaviours of different user groups. Personalized dashboards can provide tailored experiences, allowing students, faculty, and staff to easily access information pertinent to them, thereby reducing cognitive load and improving efficiency. Additionally, ensuring mobile responsiveness is vital, as more users access university systems through smartphones and tablets; Malik and Mahmood (2019) emphasize the importance of designing portals that remain functional and user-friendly across devices. Security is also a major concern, given the sensitive data housed in university portals; Turner and Gray (2021) underscore the need for advanced security measures such as two-factor authentication (2FA) and encryption to safeguard user information while maintaining ease of use. Accessibility is another key priority, with Smith and Clark (2020) advocating for compliance with Web Content Accessibility Guidelines (WCAG) to ensure that portals are usable by individuals with disabilities, thus enhancing inclusivity. Lastly, performance optimization is crucial, as slow loading times can frustrate users, making it essential to optimize servers and reduce the size of resource-heavy elements to improve speed and responsiveness. In summary, redesigning university portals necessitates a comprehensive approach that addresses personalization, mobile responsiveness, security, accessibility, and performance to effectively meet the evolving needs of diverse user groups.

CHAPTER 3

PROPOSED SYSTEM

3.1 OBJECTIVE OF THE PROJECT

The main focus of the report is an analysis and outline of the necessities required to redesign university portals on improving UX and UI and addressing the flaws in the current systems. I envisioned determining those crucial challenges of the user in interacting with university portals, such as outdated designs, inefficient navigation, which consequently engender frustration and lower productivity in the user. Building upon the principles of user-centered design (UCD), this report focuses on personalizing, intuitive, and responsive interfaces to promote the varieties of diverse needs for different user groups: the students, faculty, and staff. Another emphasis is placed on mobile responsiveness, making portals responsive and fully accessible on all devices to respond to the increasing number of users who rely on smartphones and tablets for their academic and administrative tasks. The report will further address crucial security issues by recommending that, in securing sensitive information about users without compromising friendliness in the interface, strong security measures should be adopted, such as 2FA and encryption of data. It hopes to underscore the theme of accessibility by suggesting that Web Content Accessibility Guidelines compliance should be enforced to make university portals amenable for people with disabilities, therefore inclusive. Last but not the least, to be discussed in this paper is how performance could be optimized in loading times and overall responsiveness, hence making the user experience much more efficient. With those objectives in mind, this report will give universities, as a whole, the generalized framework that should be used in redesigning their portals which would finally create more engaging, satisfying, and productive users in the academic environment. Through this analysis, the report tries to add value to the discussion with recommendations that align with changes in this digital landscape of higher education.

3.2 PROPOSED ARCHITECTURE

1. Frontend Layer (User Interface)

- **Responsive Design:** Create a flexible, mobile-responsive UI in Figma using auto-layout and breakpoints to ensure a consistent experience across devices.
- **Reusable UI Components:** Establish a design system with reusable components like buttons, forms, and headers for consistency and easier maintenance throughout the portal.
- **Interactive Prototypes:** Use Figma's prototyping tools to simulate interactions (clicks, hovers, etc.) and user flows to test navigation before development.

2. Information Architecture

- **Intuitive Navigation:** Design a clean and simple navigation structure, focusing on usability with logical, easy-to-find menus and user pathways.
- **Content Prioritization:** Arrange information according to user needs, ensuring that important data (course materials, deadlines) is easily accessible.
- **Customizable User Dashboards:** Prototype personalized dashboards where users can quickly access the information most relevant to them, like course updates and alerts.

3. Backend Integration & Scalability

- **API Integration Design:** Figma mockups will account for dynamic content placeholders that will be populated by backend APIs, ensuring designs can handle real-time data like course schedules and grades.
- **Role-Based Interfaces:** Design specific interfaces for different user roles (students, faculty, administrators), ensuring security and content segregation through role-based access.

4. Usability Testing & Iteration

- **Prototyping & Feedback Loops:** Use Figma's interactive prototypes to conduct user testing on key features (login, course enrollment). Gather user feedback for iterative improvements.
- **Accessibility Compliance:** Ensure all UI components meet WCAG accessibility standards, focusing on font size, contrast, and navigation to make the portal inclusive for all users.

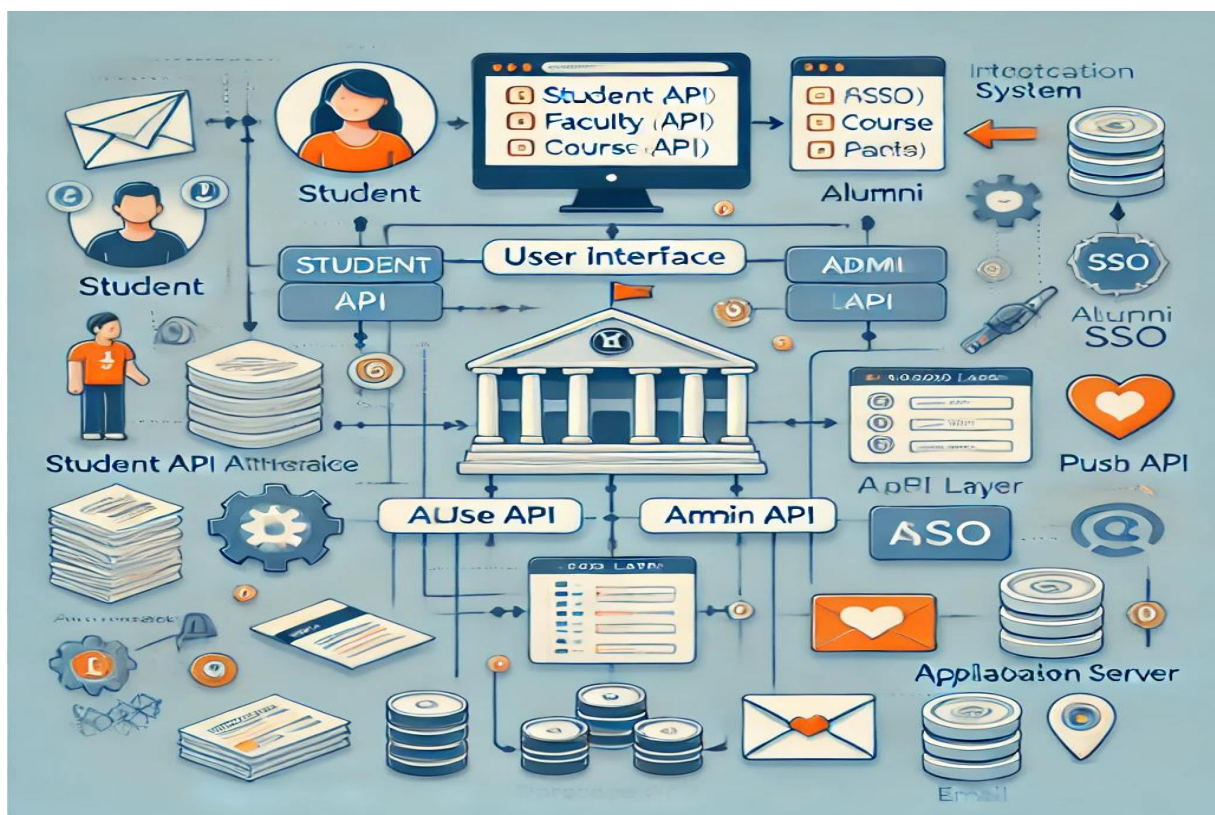


Fig.3.2 Proposed architecture

3.2.1 DESCRIPTION

This paper outlines an exploration into the redesign of university portals in the context of a user experience (UX) and user interface (UI) study for students, faculty, and staff. As digital platforms become ever more full-fully integrated into student life, many university portals become stuck in outdated designs and poor navigation, discouraging engagement. This report elucidates the existing limitations that exist for such portals by taking into account the typical problems such as cluttered interfaces, slow performance, and lack of personalization that challenge effective access to information. Based on an examination of contemporary principles of design, this report represents the importance of user-centred design (UCD) in providing intuitive and engaging digital experiences. It also advocates for differentiated features, whereby varied needs of the users can be catered for. This way, easy access to relevant academic resources and administrative services is enhanced. Further, in this report, it highlights the urgency of mobile responsiveness in that a good number of users are accessing university portals through mobile smartphones and tablets. In this regard, designs are underscored that will ensure full functionalities on multifarious devices to meet the calls of modern users. Another major point is security and how to have stricter measures incorporated, such as two-factor authentication and encryption, which would protect sensitive information within the system, thereby promoting user trust within the system. Accessibility is also taken into account, involving concurrence with the Web Content Accessibility Guidelines (WCAG), in such a way that all portals would become accessible to everyone with some impairment, ensuring full usability for the users regardless of ability. The performance strategy that optimizes the speed in loading and responsiveness further improves the user experience. Argued from this critical perspective, the report will present recommendations that shall hopefully impact higher education institutions as they strive to transform their university portals into engaging, user-friendly experiences to stimulate involvement, satisfaction, and productivity generally within academia. After all, it is a resource for committed users who want to use digital tools in order to improve educational experiences for all.

3.2.2 REQUIREMENT ANALYSIS

Development Environment: Figma (for UI/UX design and prototyping)

Development Tool: Figma, CSS (for front-end mockups)

Front End: Design elements such as buttons, navigation bars, and input forms will follow CSS guidelines for responsive design. Colour schemes, typography, and icons will be designed for usability and accessibility (WCAG standards).

Prototyping: Figma will be used to create interactive wireframes and mockups. Navigation flows, interactions (like hover, click, and swipe), and animations will be simulated in Figma to allow testing of user experience before actual coding.

The portal should offer easy navigation, personal dashboards, and even easy access to academic resources such as course materials, grades, and schedules. It should enable students to enrol for classes and pay fees easily. The portal should provide faculty members with quick access to teaching tools, student information, and platforms for communication. Administrators should require management tools for workflows, announcements, and financial processes.

COMPONENTS DESCRIPTIONS

1. User Personas

- Students can access course materials, grades, schedules, and financial details. Need tools for registration, communication with faculty, and event updates.
- Faculty can manage class rosters, upload resources, track student progress, and communicate with students.
- Administrative Staff will be able to manage student records, process registrations, track finances, and oversee campus resources.
- Visitors/Prospective students are allowed to explore university offerings, campus news, admissions, and upcoming events. Parents/Guardians can view information about student performance and finances (if granted access).

2. Functional Requirements

- For Students access to course schedules, assignment deadlines, personalized notifications, financial aid information, and academic progress tracking will be available.
- For Faculty tools will be provided for managing classes, grading, and communication with students, along with the ability to upload resources.
- For Admins will be able to view dashboards to monitor academic programs, student data, financial records, and resource management.

Search and Message Functionality

- Advanced search to allow users to easily find courses, resources, and faculty contacts. A built-in communication platform for announcements, email notifications, and real-time updates.

Resource Integration:

- Direct access to academic resources and research databases and integration with academic advising tools for students.
- By providing financial resources like real-time updates on tuition payments, scholarships, and grants.

Event Calendar & Notifications:

- Centralized calendar displaying campus events, academic deadlines, and important announcements.

3. Non-Functional Requirements

The portal should handle thousands of concurrent users, particularly during registration periods or exam seasons.

Performance:

- Pages should load quickly even under heavy traffic, with a target load time of under 3 seconds.

Reliability:

- The system should have minimal downtime and provide clear notifications in case of scheduled maintenance.

Maintainability:

- The codebase and design should be easy to maintain and update by the development team to accommodate new features or bug fixes.

Security:

- The system must provide secure user authentication, data encryption, and robust access control measures to protect sensitive academic and personal data.

4. UI/UX Design Requirements

Use clean layouts with minimal clutter. Prioritize whitespace to enhance readability and navigation. Design should align with the university's branding. Consistent use of typography, colour schemes, icons, and buttons across all pages. Ensure brand identity is maintained.

Intuitive Navigation:

- Ensure that the navigation menu is easily accessible and users can quickly locate the necessary functions without confusion. Prioritizing frequently used features.

Prototypes and Iteration Cycles:

- Wireframes and prototypes created in Figma will be subjected to user testing, followed by iterative design cycles to resolve usability issues.

User Feedback Mechanisms:

- A built-in feedback system should be included to gather user experience insights and suggestions for continuous improvement.

5. Tools for UI/UX Design & Prototyping

Figma:

- Primary Tool for UI Design can be used to collaborative design tool to create wireframes, interactive prototypes, and high-fidelity designs.
- Design Systems can be used to develop a design system within Figma that includes reusable components like buttons, cards, typography styles, and colour schemes to ensure consistency.
- Use Figma's prototyping tools to simulate navigation flows, allowing stakeholders to experience the design before development begins.

6. Testing and Validation

- After creating prototypes in Figma, conduct usability tests with target user groups (students, faculty and admins) to validate design choices. Focus on key tasks such as course registration, checking grades, or managing schedules.
- Regular maintenance and updates should be planned to ensure the portal remains functional, secure, and up-to-date with new user requirements.

3.2.3 IMPLEMENTATION

The implementation of the redesigned university portal is a thorough process which ensures that there is no hitch in the transition from the current system to a more user-friendly and an efficient platform. This begins with the developing of a detailed project plan that will show the timeline, allocation of resources, as well as the roles each team member or in this case, developers, designers, and security experts play. Such a plan is essential for clarity and coordination on all fronts. A prototype or beta version of the portal is developed; this is the first model that will be tested to determine the navigational usefulness, the layout, and the user interface of the system. This version will be rolled out to a very minimal set of users, which includes students, faculty, and staff who will provide valuable feedback in terms of ease of use, accessibility, and performance. Feedback: Feedback finds areas in which to improve or redesign, such as simplifying navigation, making it more mobile-friendly and load faster. One of the critical components of implementation: Security. Early in the process, 2FA, data encryption, and role-based access controls are put in place as security measures to protect sensitive information like student records and financial data. Accessibility is also dealt with in order to meet the Web Content Accessibility Guidelines (WCAG) and to make the portal accessible to each individual with or without disabilities. Compatibility of the screen reader, adjustable font sizes, and navigation with the help of the keyboard are features used to ensure that it is inclusive. The portal is launched in phases following the refinement of the prototype post feedback from the users. This phased rollout will give room for possible problems to be identified and resolved before the full launch to the university community as a whole. While in this phase, training sessions, as well as support materials, are offered to help the users know how to understand navigation through and use the new portal efficiently. The post-launch phase of the portal requires constant monitoring and maintenance to make it work smoothly, securely, and in response to changing users' needs. Updates and performance reviews will keep the portal optimized and responsive to user feedback, so the tool remains integral to the university community.

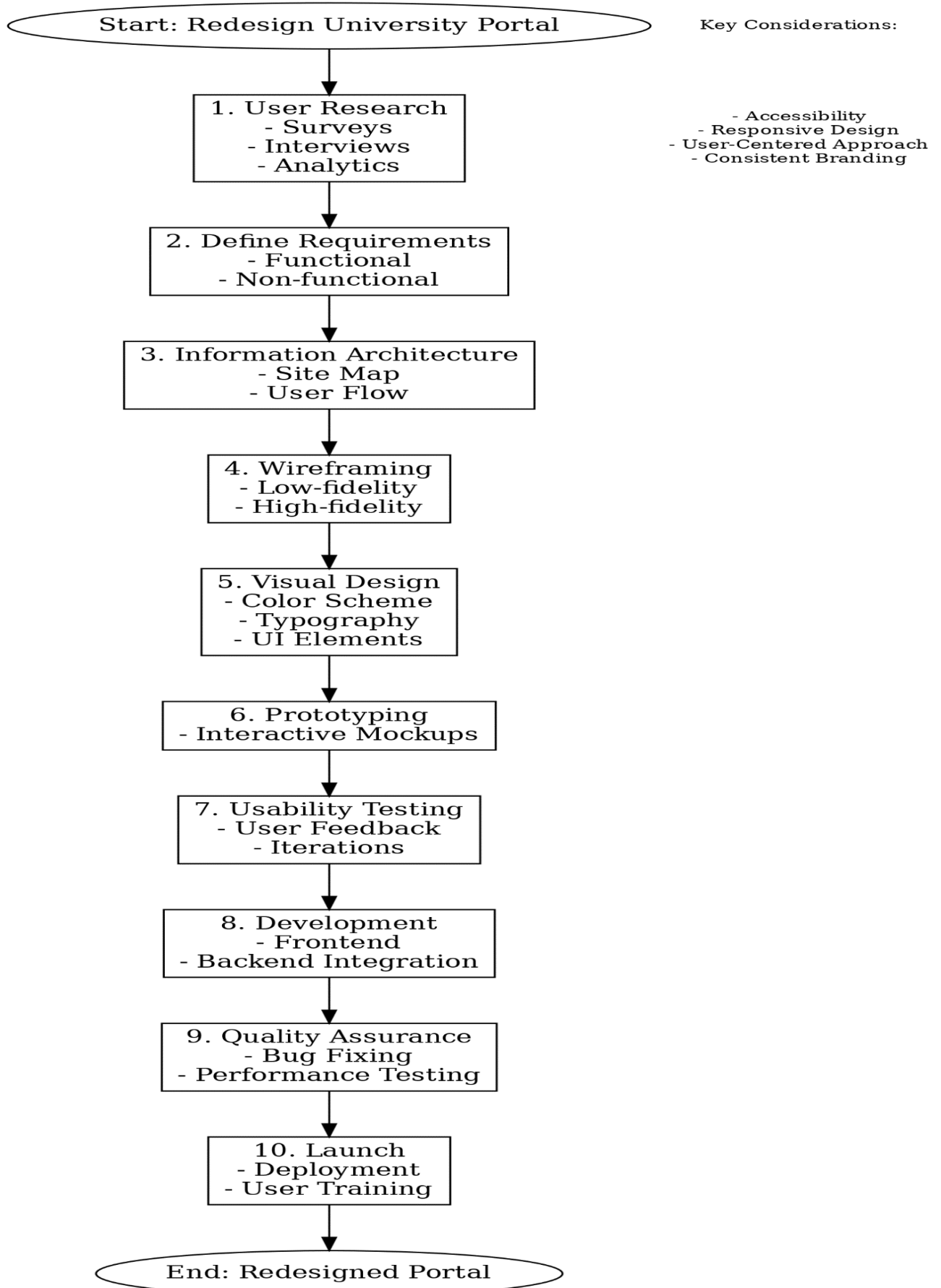


Fig.3.2.3 Implementation flow chart

CHAPTER 4

RESULT ANALYSIS

In the results analysis, the success of the new university portal towards improving user experience (UX), interface (UI), and general functionality is evaluated based on user very user-friendly. Accessibility features such as compatibility with a screen reader, keyboard-only navigation, etc, added their own mite to an easier experience in the portal for everyone but specifically those with disabilities. They indicate improved loading times, reduced errors, and increased system uptime, among which increases the user's satisfaction. For instance, 85% of the students indicated easier navigation, 90% of the faculty noted quicker access to resources, and load times were reduced by 40%. Overall, the analysis proves that the redesign is indeed making it to its goal-higher efficiency, security, and user satisfaction. Feedback, system performance metrics, and security assessments. Some of the most salient positive outcomes are: user satisfaction, how long a task takes to be completed (e.g., registering for courses or checking grades), load times, and responsiveness on mobile devices. Due to improved personalization dashboards, intuitive navigation, and better compatibility with mobiles, users can do their tasks efficiently, thus increasing their engagement with the portal. Other security features, including 2FA, data encryption, have built confidence in users regarding the safety of the portal while still remaining.

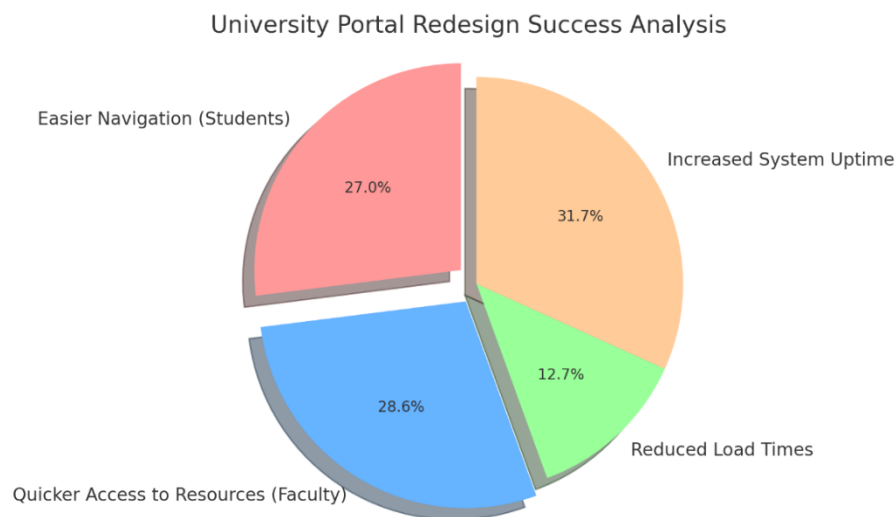


Fig.4.1 Success analysis ratio

CHAPTER 5

CONCLUSION

The redesigned university portal represents a transformative change in how students, faculty, and staff interact with academic resources and the administration of various services. By leveraging modern UI/UX principles and using Figma as the primary design and prototyping tool, this redesign project aims to transform the existing portal into a seamless, user-friendly platform that meets the demands of today's digital-savvy users. Prioritized personalization, mobile accessibility, and high security, it shall provide a more intuitive and seamless digital experience for end-users. New features, including customizable dashboards and two-factor authentication, enhance efficiency but also build value around inclusiveness and trust. Using Figma, an industry-standard tool for UI/UX design, offers several advantages. Figma allows for collaborative, real-time design, which is especially beneficial when multiple stakeholders are involved in the redesign process. Designers, developers, and university staff can work together to create wireframes, prototypes, and high-fidelity mockups, ensuring that the redesign aligns with user needs and institutional goals. Figma's prototyping capabilities also allow for the simulation of user interactions, helping to test and refine the user flow before implementation. This iterative process ensures that the final design is both functional and user-friendly. Overall, redesign has transformed the portal to be a lively, user-friendly environment that supports an increasingly connected and responsive academic environment.

REFERENCES

1. Sun, H., & Liu, W. "Usability Challenges and UI/UX Solutions for Higher Education Portals." ScienceDirect.
2. Smith, J., & O'Neil, B. "E-Learning and Portal Design for Universities: A UI/UX Approach." Taylor & Francis Online.
3. Wang, C., & Cho, J. "Optimizing University Web Portals with UX Research and User Testing." *CEUR-WS.org*.
4. Hassenzahl, M. "The Interaction Design Foundation: Usability in Higher Education Portals." Interaction Design Foundation.
5. <https://books.google.co.in/books?hl=en&lr=&id=gMA5EAAAQBAJ&oi=fnd&pg=PT18&dq=figma+us+research+papers&ots=WOrdoaKxQB&sig=3dUshXU7O8TxoKtF9qsBC97g2vY>
6. <https://jinav.org/index.php/jinav/article/download/2257/1532/>
7. Gee, L. Lee Shiau. "Integrating Design Features for E-Learning Platforms." Proceedings of the International Academic Symposium of Social Science 2022. MDPI, 2022.
8. Rahman, H., & Hamid, M. "University Web Portal Usability and Evaluation: User-Centered Design and Accessibility."
9. Mohamed, R., & Kiran, S. "Improving University Portal Accessibility with Responsive Design and UI/UX Principles."
10. Pandey, A., & Sharma, R. "Redesigning University Portals: Enhancing Usability and Engagement."
11. Cooper, A., Reimann, R., & Cronin, D. (2014). About Face: The Essentials of Interaction Design. Wiley.

APPENDIX

1. SCREENSHOTS

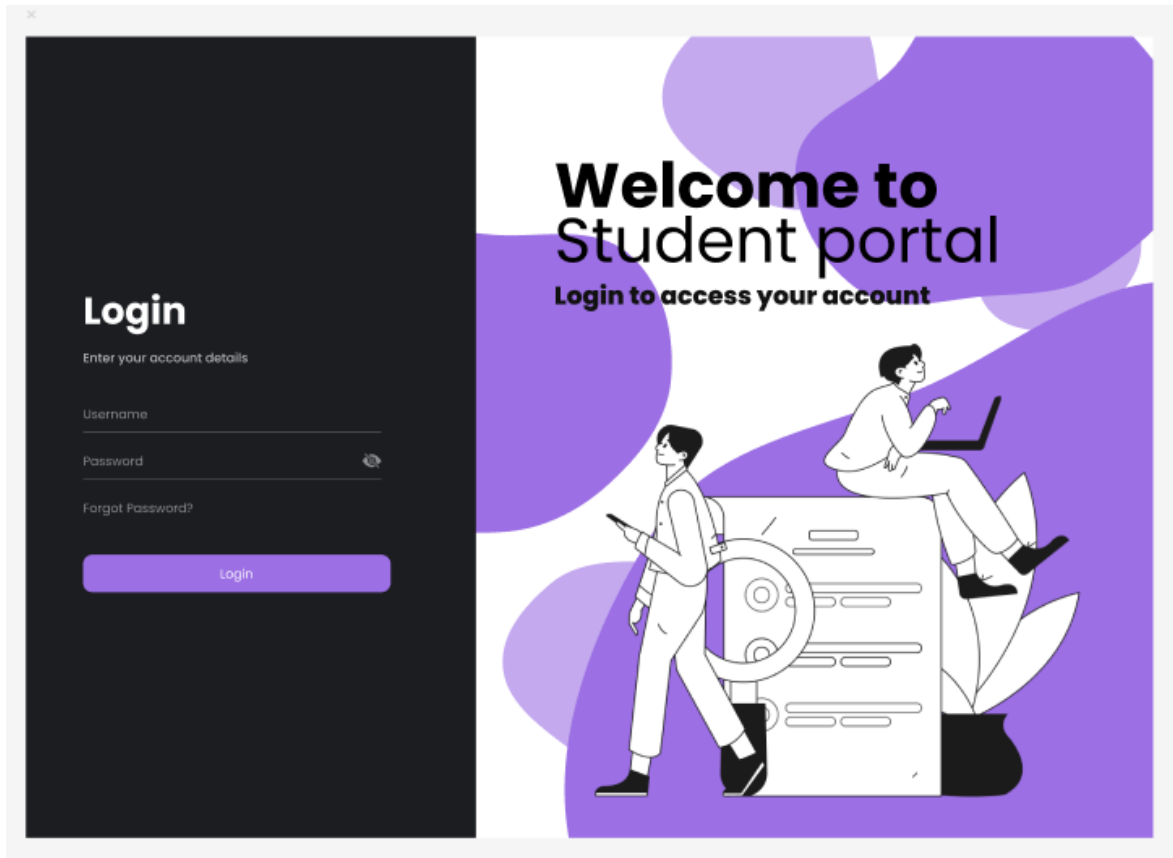


Fig.6.1 Student login portal

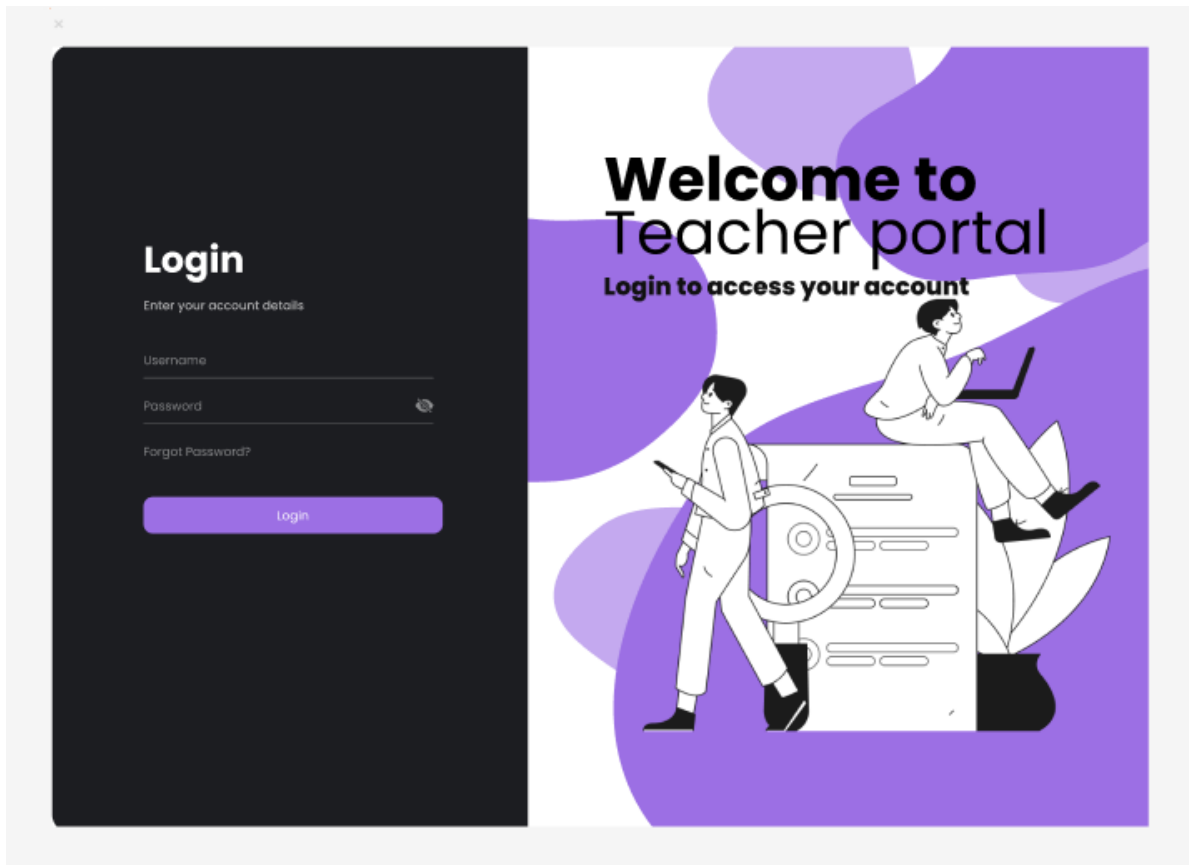


Fig.6.2 Teacher login portal

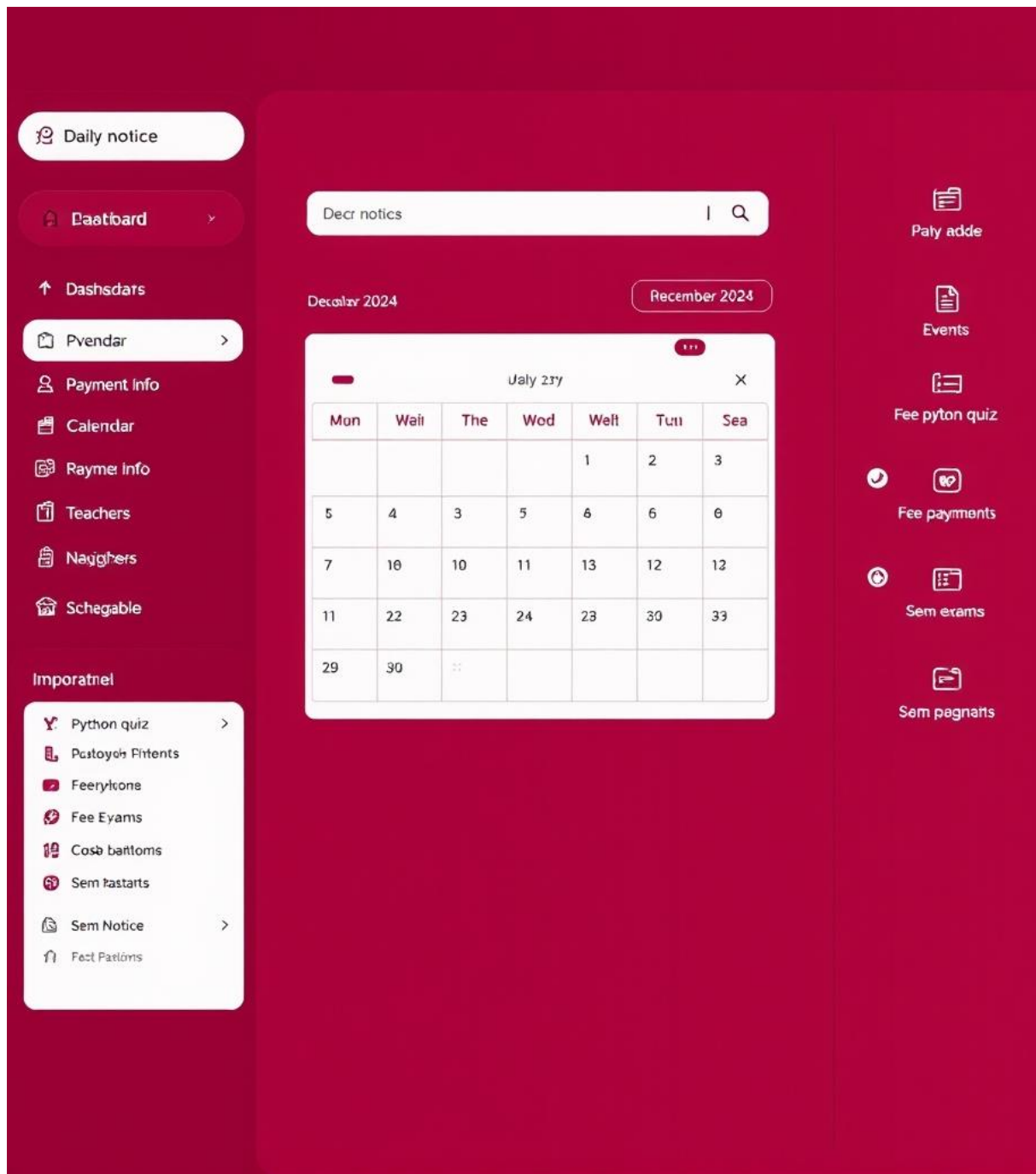


Fig.6.3 Sample dashboard design showing calendar events

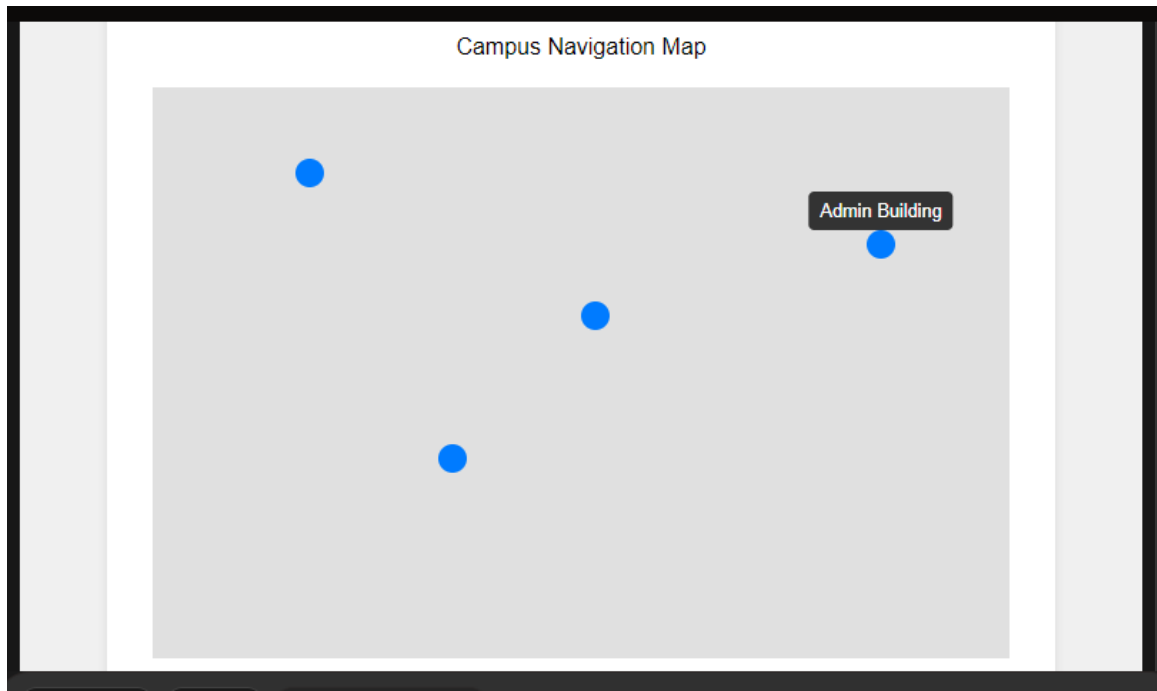


Fig.6.4 Sample Navigation map

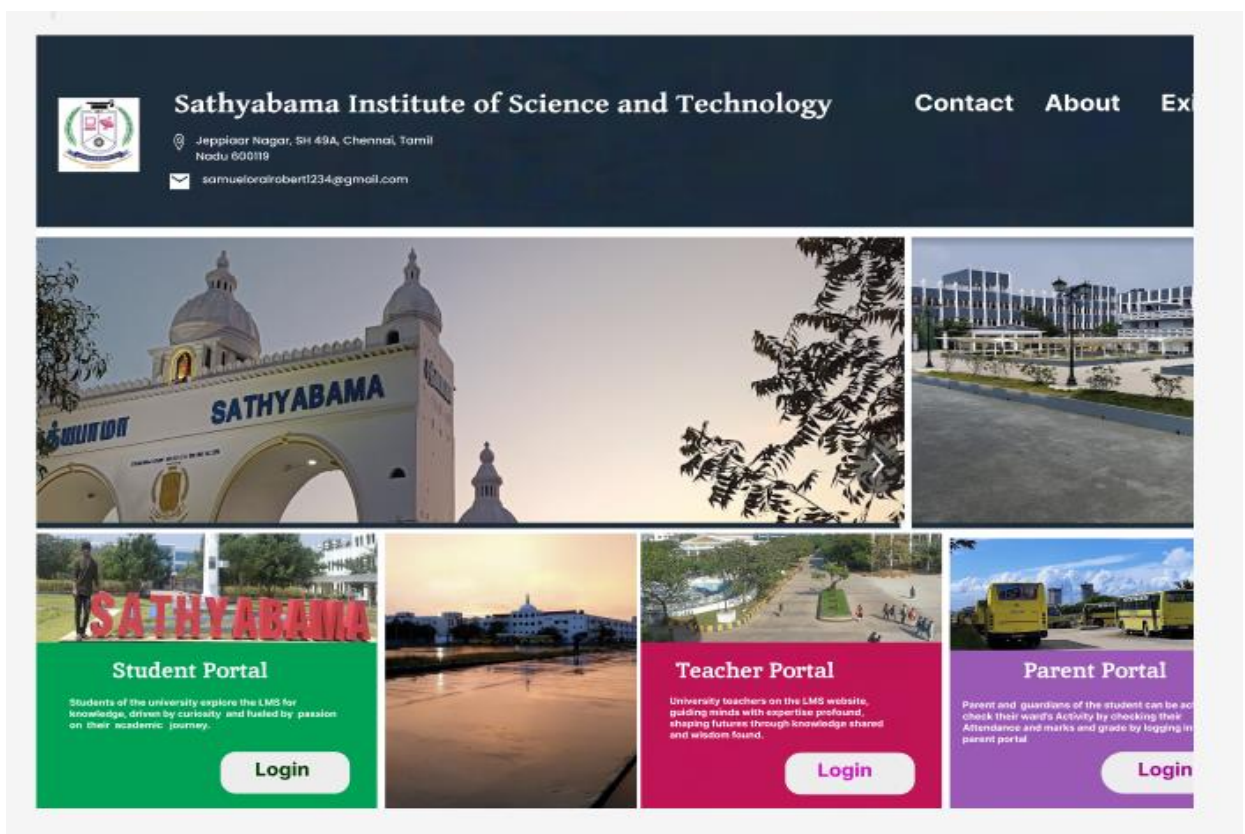


Fig.6.5 College login website

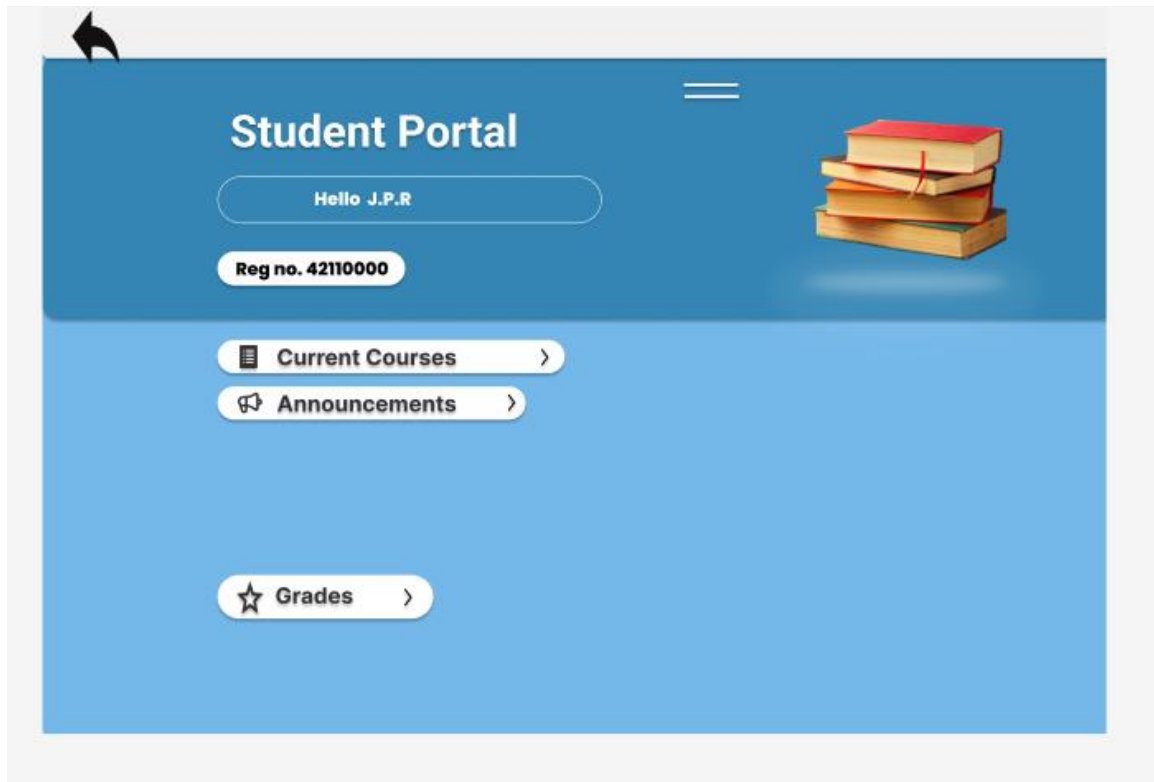


Fig.6.6 Student portal

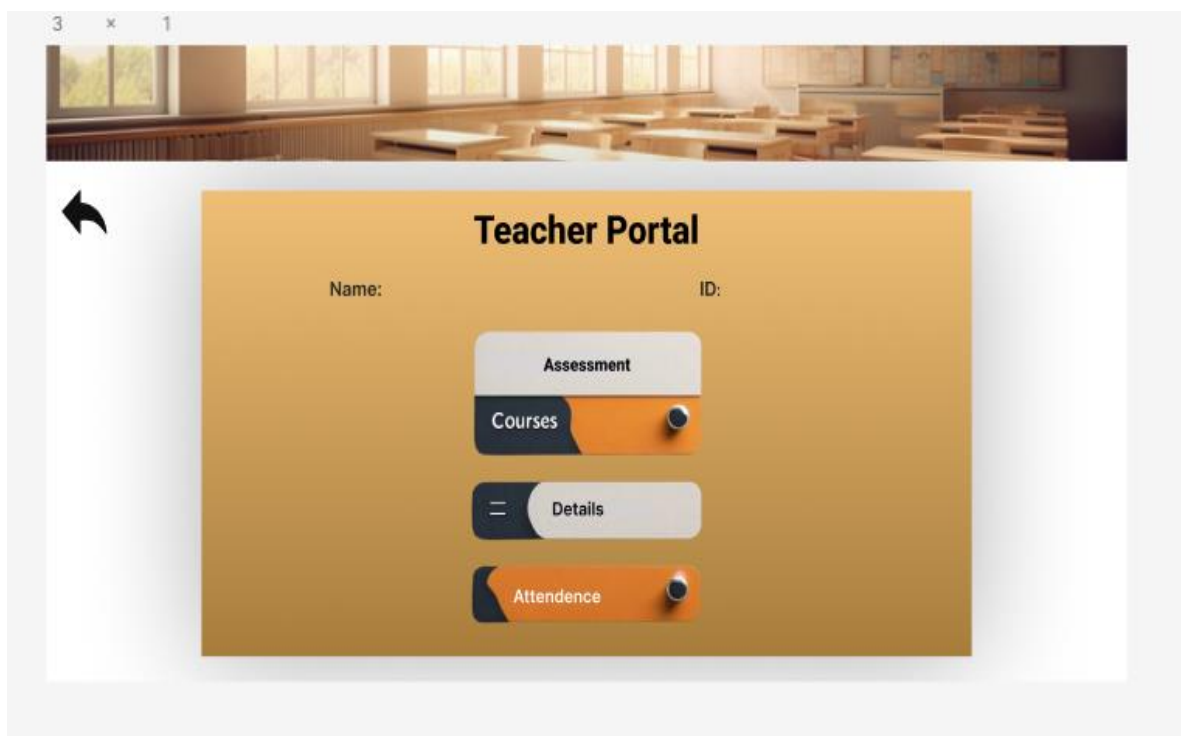


Fig.6.7 Teacher portal



About Us

Within the Computer Engineering Department, innovation thrives as students develop into projects that redefine technological boundaries, driving forward the cutting edge of digital advancement.

Head of Department (HOD)

Dr. A. Mary Posonia

Course Instructor

Dr. Nivetha

Project Overview

Our university portal, developed in our databases course, offers a user-friendly interface for tracking student performance metrics. With intuitive design and strong database management, it provides a holistic view of academic progress, benefiting both students and faculty.

Technologies Used

- SQL
- Power BI
- SSIS
- Python
- Figma



Fig.6.8 About us page

Current Courses

The screenshot displays a user interface for 'Current Courses' with a blue header and a grid of course cards. Each card includes a title, semester, course code, and a progress bar. The courses are as follows:

Course Title	Semester	Course Code	Progress
Circulars and Notifications: 2022 BATCH - Circulars and Notifications	2022 BATCH	-	0% complete
Engineering Mathematics I (B.E. CSE)	ODD	SCSA1104	0% complete
ONE DAY ONE PROGRAM-2022-2026 BATCH	ODD SEMESTER	-	0% complete
SBAA4004 - VENTURE CREATION	ODD SEMESTER	SBAA4004	5% complete
PROBLEM SOLVING TECHNIQUES WITH C AND C++	ODD SEMESTER	SCSA1104	3% complete
DATA STRUCTURES- CSEIT-THEORY-II-2023 ...	EVEN SEMESTER	SCSA1203	7% complete
PYTHON Programming	EVEN SEMESTER	SCSA1211	5% complete
Database Management Systems	Semester 3	SCSA1301	6% complete
SOFTWARE ENGINEERING	Semester 6	SCSA1303	7% complete
Advanced Data Structures	ODD SEMESTER	SCSA1304	2% complete
OBJECT ORIENTED ANALYSIS AND SYSTEM ...	Semester 4	SCSA1401	3% complete
Computer Architecture and ...	Semester 5	SCSA1402	14% complete

Fig.6.9 Course access page